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Problem 1. Solve the radial wave equation on a circular membrane with $c = 1$, $R = 1$, zero initial velocity, and initial displacement $u(r, 0) = f(r)$, where

$$f(r) = 1.108J_0(2.4048r) - 0.140J_0(5.5201r) + 0.045J_0(8.6537r).$$

Additionally, find all the harmonic frequencies and the radius of each of the (inner) nodal circle for each mode (m) of the solution.

For reference:

$$\alpha_1 = 2.4048, \quad \alpha_2 = 5.5201, \quad \alpha_3 = 8.6537, \quad \alpha_4 = 11.7915, \quad \alpha_5 = 14.9309$$

[10 marks]

Solution.

(Solution continued)

□

END OF QUIZ 9